

DUSHIMIMANA VALENTIN

Smart Computing Student · AI / Explainable AI (XAI) Researcher · Full-Stack Developer

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PROFILE

Senior Smart Computing student at Kyungdong University with a focus on artificial intelligence, explainable AI (XAI), computer vision, and full-stack development. Published SSRN research on explainable AI for non-expert users. Currently building MIZERO, an AI-powered conversational lost-and-found platform for Rwanda, as a graduation project. Seeking to pursue a Master's degree abroad in embedded systems / AI starting Fall 2027.

EDUCATION

Bachelor's Degree in Smart Computing

Kyungdong University, Sokcho, South Korea — 2024 – Present (Senior)

High School Diploma

Master Dei High School, Rwanda — 2019 – 2023

PUBLICATIONS

"Explainable AI for Non-Expert Users: A Systematic Mapping Study Using the Petersen et al. SMS Approach"

SSRN Working Paper — Abstract ID: 7436539

A systematic mapping study reviewing how Explainable AI (XAI) research addresses the needs of non-expert users, applying the Petersen et al. systematic mapping methodology to identify gaps and trends in the literature.

FEATURED PROJECT

MIZERO — AI-Powered Lost & Found Platform for Rwanda

Graduation Project — Full-Stack (Next.js), in progress, target completion December 2026

- Designed and building a full-stack conversational lost-and-found and intelligent matching platform for Rwanda, for both web and mobile.
- Built an image classifier (transfer learning on MobileNetV2) to auto-tag lost-item photos by category — ~98% validation accuracy on 3,500+ images.
- Built an NLP intent classifier (TF-IDF + Logistic Regression) to interpret user messages in the platform's chat interface.
- Upgrading the chatbot to a full conversational AI (OpenAI API, text and voice) for a more natural user experience.

OTHER PROJECTS

Explainable Credit Risk Classifier — Random Forest credit risk model explained with global and local SHAP visualizations.

Mobile Money Fraud Detection — Random Forest classifier detecting account-draining fraud patterns; 88% recall on a synthetic dataset.

Blockchain Asset Registry — Solidity smart contract for a tamper-proof land/asset registry with ownership transfer and independent verification.

Computer Vision & Image Processing — Classical CV techniques (thresholding, Sobel, Canny, morphology) using Python, OpenCV and Scikit-image.

DUSHIPLAT Adventure (Unity Game) — Touch-based 2D mobile game built in Unity/C# with 5+ levels and Android controls.

Car Parking Management System — Python/Spyder desktop GUI application for managing vehicle entry, exit, and parking records.

SKILLS

Programming Languages	Python, Java, C#, C/C++, JavaScript, SQL
Web & Frameworks	HTML, CSS, React, Node.js, Next.js, Django
AI / ML	Computer Vision, NLP, Explainable AI (SHAP), Transfer Learning, Data Mining
Tools & Platforms	Git/GitHub, Docker, VS Code, MySQL, MS SQL Server, Linux, VirtualBox
Other	Robotics (Arduino), Cloud Computing, Cybersecurity, Data Structures & Algorithms

ACHIEVEMENTS & GOALS

- Published research on SSRN as part of graduate school application preparation.
- Building real-world AI and full-stack projects spanning computer vision, NLP, and explainable AI.
- Pursuing a Master's degree abroad (US/Canada) in embedded systems / AI, targeting Fall 2027.
- Contributing to Rwanda's digital transformation through the MIZERO platform.